

Naïve Bayes Classification

Experiment:

Breast Cancer Analysis and Naïve Bayes Classification

Using the Breast_Cancer.csv dataset:

1. Perform basic EDA by displaying the first five records, missing values, descriptive statistics, and diagnosis distribution.
2. Plot a scatter plot of **radius_mean vs texture_mean**, a histogram of **radius_mean**, and a box plot of **radius_mean by diagnosis**.
3. Apply **Gaussian Naïve Bayes** to predict diagnosis using the numerical features, with an **80:20 train-test split**.
4. Display the **classification report and accuracy**.
- 5.

Experiment:

Iris Flower Analysis and Naïve Bayes Classification

Using the Iris.csv dataset:

1. Perform basic EDA by displaying the first five records, missing values, descriptive statistics, and species distribution.
2. Plot a scatter plot of **Sepal Length vs Sepal Width**, a histogram of **Sepal Length**, and a box plot of **Sepal Length by Species**.
3. Apply **Gaussian Naïve Bayes** to predict Species using the numerical features, with an **80:20 train-test split**.
4. Display the **classification report and accuracy**

Experiment:

Wine Quality Analysis and Naïve Bayes Classification

Using the Wine_Quality.csv dataset:

1. Perform basic EDA by displaying the first five records, missing values, descriptive statistics, and quality distribution.
2. Plot a scatter plot of **alcohol vs pH**, a histogram of **alcohol**, and a box plot of **alcohol by quality**.

3. Apply **Gaussian Naïve Bayes** to predict quality using the numerical features, with an **80:20 train-test split**.
4. Display the **classification report and accuracy**.

complete all Naïve Bayes programs Before 23/09/2026